

Analysis of RNA-Protein complexes predicted with deep learning

University of Freiburg



Master Project Report

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Views and opinions expressed are those of the author(s) only.

Abstract

This work explores the use of deep learning models for predicting RNA–protein interactions (RPIs). We observed the performance of an RPI classifier. We prepared a RPI dataset with the RNAcompete methodology for observing how AlphaFold3 differentiates RPIs within some of its confidence metrics. We used TabPFN to classify RPIs by also using the prepared dataset for in-context learning at inference time. Results demonstrate that the classifier might still need more generalization capabilities, AlphaFold3 can recognize sequence level signals of RNA-protein binding and TabPFN can succeed at RPI recognition given the right feature set.

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Chapter 1

Introduction

The driving biological actors of this work are the RNA-Protein Interactions (RPIs), as their hold tremendous importance in life.

In light of the recent thriving of deep learning driven prediction tools for molecules alone, the next big step in this domain would mean modeling interactions between them. However, realizing such a task is significantly more challenging due to the complexity of binding folding and dynamics.

In the following work, we focus on using and evaluating domain-specific deep learning models which can help to describe and understand RPIs by modelling and classifying them.

The term "molecules" will also refer to either protein residues, protein aminoacids or RNA nucleotides; to denote biological entities at a greater scale than atoms.

The following concepts were of great use for the evaluations made:

1.1 AlphaFold3

AlphaFold3 [1] is the latest version of DeepMind’s protein structure prediction system, now expanded to model not only individual proteins but also protein–protein, protein–RNA, and other biomolecular complexes. Recently, it has gained the attention of the community for its accurate predictions on proteins alone. However, it is believed that its modeling capabilities still lack accuracy regarding some other bio-molecules and multi-molecular complexes, and therefore, more caution must be put when working with these.

In this work, AlphaFold3 was considered as a structural resource to iden-

tify points suggesting RPI within the sequences. Also, to get an impression on how correct its predictions can be on RPIs.

1.1.1 PAE

PAE (Predicted Aligned Error) is a metric of relative positional confidence between pairs of molecules (protein residues, RNA nucleotides, etc.) from the predicted structure. More specifically, $PAE(A,B)$ tells how much misplaced (expected positional error in Angstroms) a molecule B is with respect to the frame of reference of another molecule A, and this is calculated from each molecule’s frame of reference (so the other case is $PAE(B,A)$ for this pair of molecules). The errors can differ slightly between the two cases as a result of possible disordered regions involving any of the two molecules.

The PAE is provided in a square matrix with dimension number of residues plus number of nucleotides in an AlphaFold3 output JSON file.

1.1.2 pLDDT

pLDDT (predicted Local Distance Difference Test) is a per-atom metric of local confidence. It is scaled from 0 to 100, with higher scores indicating higher confidence. It measures the expected local accuracy for the atoms of a given molecule. The pLDDT is provided in a one dimensional array with dimension number of atoms in an AlphaFold3 output JSON file.

1.2 Distance matrix

In several parts of this work, we leverage AlphaFold3 RPI complex predictions by creating a distance matrix between all of their protein residues and RNA nucleotides.

This matrix (with dimensions number of residues by number of nucleotides) is a crucial component. From it, residue - nucleotide pairs being within five Angstroms from each other are identified, which can imply potential RPI points.

A cell in the matrix represents the shortest possible distance, in Angstroms, for a given residue-nucleotide pair. The distance is taken as the simple euclidean distance between the closest two atoms in the pair. For atom selection, all non-heavy atoms (i.e. Hydrogen atoms) are ignored.

The original matrix shape of the confidence metrics had to be adapted to match the molecule-wise format of the distance matrix.

The figure 1.1 provides the graphical representation of this matrix, with a colorheat map for the distances and the red dots indicating residue-nucleotide pairs at five Angstroms or less from each other.

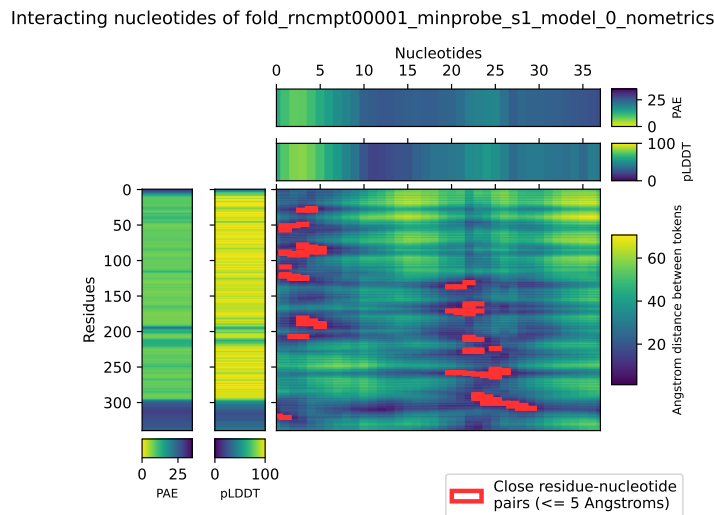


Figure 1.1: Distance matrix between residues and nucleotides of a RPI example

For example, there seems to be close enough pairs between the first five nucleotides and the first fifty residues in this RPI example.

Also in figure 1.1 the pLDDT and PAE confidence metrics are plotted to the left and upper sides of the distance matrix, likewise with a color heatmap, to illustrate what the prediction confidence is throughout the residues and nucleotides.

The close pairs later on are exported into a CSV file to carry out further evaluations on them.

1.3 RNACompete: a powerful methodology to derive RPI positives and negatives examples

RNACompete [6] is a high-throughput experimental platform designed to systematically measure the binding preferences of RNA-binding proteins (RBPs) against synthetic RNA sequences. It provides quantitative binding affinity scores between a given protein and numerous RNA sequences.

We utilized the RNACompete dataset to generate positive and negative examples required for the evaluations in this work. Specifically, we selected the protein from the RNACompete dataset and extracted six RNA-protein interaction examples: the three RNA sequences with the highest binding energies (indicating strong interaction, labeled as positive) and the three with the lowest binding energies (indicating weak or no interaction, labeled as negative). In all six cases, the protein sequence remained constant, while only the RNA sequences varied.

We did this on 244 RNACompete proteins, to get a total of 1464 RPI examples.

This approach allowed us to create a balanced and biologically meaningful dataset of interaction data.

1.4 RPIembeddor

RPIembeddor is a novel transformer-based model designed for classifying ncRNA-protein interactions [4]. The input for this are two sequences: one protein aminoacid sequence and one RNA nucleotide sequence. Both of them must be at most 1022 characters in length. The program asks the user for each of the sequences and then predicts whether they both are interacting by printing “POSITIVE INTERACTION” or “NEGATIVE INTERACTION” in the console.

1.5 bedtools

“bedtools” [5] is a suite of methods for a wide-range of genomics analysis tasks. It allows to intersect, merge, count, complement, and shuffle genomic

intervals from multiple files in widely-used genomic file formats such as BAM, BED, VCF.

The specific method we used was “getfasta”.

With this, we extracted DNA sequences from a reference genome based on coordinates specified in a BED file. For example:

Input:

A BED file containing genomic regions (e.g., chromosome, start, end).
A reference genome file in FASTA format.

Process:

The tool reads the coordinates from the BED file.
It extracts the corresponding DNA sequences from the reference genome.

Output:

A FASTA file containing the extracted sequences,
with headers derived from the BED file.

1.6 FASTA

A FASTA file is a text format used to store nucleotide or protein sequences. It consists of one or more entries, each starting with a header line (preceded by >), followed by the sequence itself. The header typically contains an identifier and optional descriptors.

1.7 TabPFN

TabPFN (Tabular Prior-Data Fitted Network) [2] is a pretrained transformer-based model designed for tabular data classification. It requires no training on the target dataset and can directly perform inference by leveraging prior knowledge learned from millions of synthetic tasks. In this project, TabPFN was used as an additional model to classify RNA–protein interactions (RPIs) as positive or negative. RPI examples were encoded into feature tables, and TabPFN was applied to evaluate its performance in predicting interactions,

providing a deep learning-based benchmark without the need for task-specific training.

Chapter 2

Methods

2.1 Examining RPIEmbeddor performance on RPI examples

The direct approach to perform this testing was a script to iterate through a set of protein-RNA pairs, running the RPIembeddor on every pair.

However, we also questioned what effect subsequences from the whole RNA could have on the RPIembeddor’s prediction. It is reasonable to think that the prediction should remain unchanged if one interacting region was found, even if considering more irrelevant nucleotides.

So our approach required that, in addition to the whole RNA, the tool was run on iteratively extended subsequences. This was only done with the RNA.

As a result, a pre-processing script was done to adapt the RNA sequences depending on whether they needed to be considered fully, partially, or with this iteratively extending routine. For this last case, “bedtools” was useful.

The general algorithm for this script was:

1. Input a file with the protein - RNA pairs and a FASTA file only with the RNA sequences.
2. Specify the adaptation for every RNA sequence and carry it out. More details on appendix A.1
3. Map the adapted RNA sequences with their corresponding protein sequence and print this to a resulting file.

4. Run the RPIembeddor on every pair of the resulting file to obtain the predictions and their associated prediction probabilities.

2.2 Comparing the AlphaFold3 output metrics distributions between RPI positives and negatives examples

Obtaining the models predicted by RPI AlphaFold3 for the dataset was necessary to carry out the evaluations.

The dataset was provided to AlphaFold3 in JSON text files containing all the RPIs examples (RNA-protein pairs), identifiers, and a random seed (1 for all examples). Once we obtained the predictions, we ran a script to create the distance matrix and obtain the close residue-nucleotide pairs in a CSV file.

Diverse plot creation scripts were then run on these CSV files.

2.3 Classifying RPIs with TabPFN and AlphaFold3

2.3.1 Creation of dataset to feed it to TabPFN

From the tables containing the close residue - nucleotide pairs obtained with the script of the distance matrix, the data for TabPF was created.

These tables contained also AlphaFold3 metrics and other structural features (like the whole protein and RNA sequences) associated with every close pair.

Since there can be more than one close pair per example in this table, the close pairs were grouped by the example of RNACompete they belonged to, and then aggregation for every AlphaFold3 metric value was performed.

Also, since TabPFN only accepts numerical features, columns missing this format had to be reshaped (encoded) like the RNA and protein sequences.

Protein sequence column was encoded into aminoacid composition columns (one column per each aminoacid count in the sequence normalized by the sequence length).

Likewise, the RNA sequence column was encoded into nucleotide composition columns.

The target column was called interaction and had only two classes, 1 for positive and 0 for negative, depending on whether they belonged to the positives or negatives in RNACompete. This was split to create a training and testing set for TabPFN.

The positives examples were as numerous as the negatives, so the datasets were balanced.

By framing this as a classification task, we trained TabPFN to distinguish interacting from non-interacting RPI examples using AlphaFold3’s prediction metrics and structural features in a table-like format. A strong TabPFN performance leveraging these features would indicate that AlphaFold3’s outputs effectively capture interaction patterns.

Chapter 3

Examining RPIEmbeddor performance on RPI examples

3.1 TSfam: RPIembeddor test set

We rerun the test set TSfam of the RPIembeddor to validate results and obtained a worse performance than expected with regards to RPI classification:

Table 3.1: Rerun of test set on RPIembeddor

Metric	Value
Accuracy	0.4920 (6160/12520)
Precision	0.3953
Recall	0.5638
F1 Score	0.4647

The reported test performance of RPIembeddor on TSfam from its paper is shown in table 3.2.

3.2 Validated small dataset of RPIS

Likewise, the performance for a validated small set of RPIS is shown in table 3.3.

Also on this validated small set, we applied the iteratively extending routine (mentioned in section 2.1), from index 0 to the right of the RNA

Table 3.2: Reported RPIembeddor test set performance

Metric	Value
Accuracy	0.667
Precision	0.550
Recall	0.627
F1 Score	0.586

Table 3.3: Validated set of RPIs performance

Metric	Value
Accuracy	0.3824 (13/34)
Precision	0.3333
Recall	0.4000
F1 Score	0.3636

sequence, such that we could observe the impact of sequence length on the predictions. Some of the RPI examples changed their prediction when considering longer sequences, as we can see in figure 3.1.

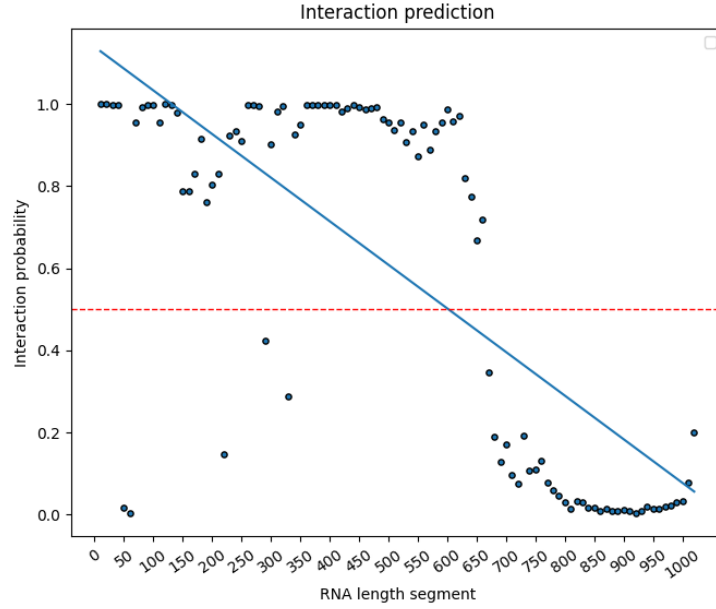


Figure 3.1: Change in prediction with sequence length

This behavior was interpreted as RPIembeddor refining its predictions since more information on the nucleotide sequence was provided. The inclusion of more nucleotides involves possibly a change in the RNA folding and the complex binding dynamics, crucial for interaction to happen, therefore having a decisive effect which can lead to change in the prediction. This behavior was not considered to be wrong.

In contrast, there were also some examples which didn't varied its prediction, even though more and more nucleotides were iteratively included, as in figure 3.2 (also RNA iteratively extended from index 0 to the right).

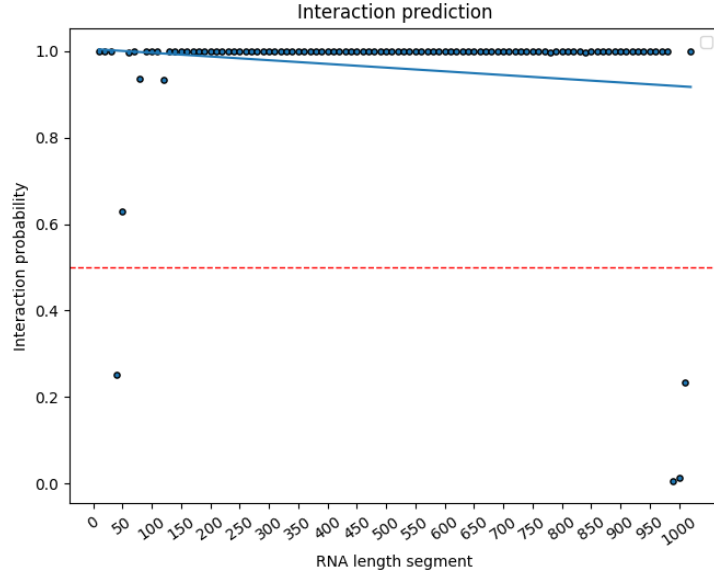


Figure 3.2: No change in prediction with sequence length

Chapter 4

Comparing the AlphaFold3 output metrics distributions between RPI positives and negatives examples

4.1 Evaluation Results

4.1.1 Datasets

In order to do the comparison, two datasets of positives and negatives RPI examples were utilized. The first was the RNACompete derived dataset mentioned in section 1.3. This consisted of 730 positives and 732 negatives.

For the second dataset, which we will refer to as the “alternative dataset”, a set of RNA-binding proteins from the Protein Databank on which AlphaFold3 was trained, was regarded as the positives. In this case, another smaller RNACompete derived dataset was used as the negatives, by extracting the RNA–protein interaction example with the lowest binding energy.

The resulting alternative dataset consisted in 221 positives and 229 negatives.

The following discussion will only focus on the plots from the RNACompete dataset. The alternative one will be in the appendix A.2 section.

4.1.2 AlphaFold3 Confidence in RNA-Protein Interactions

We compared the AlphaFold3 outputs (pLDDT, PAE and probability of residue/nucleotide contacts) distributions between positives and negatives RNA-protein pairs in both datasets. All AlphaFold3 outputs distribution were similar, suggesting AlphaFold3 does not strongly distinguish interactions using this. (Figures 4.1 - 4.3)

Specifically regarding the PAE, similar distributions for positives and negatives were not expected, as AlphaFold3 documentation states that “if two tokens have a low PAE and belong to different entities, like a protein and a nucleic acid, it suggests there is an interaction between those parts of the entities”. From this, one would have expected consistently lower PAEs on the positives examples rather than the negatives.

Meanwhile, the distributions of these outputs were significantly more confident for the alternative dataset, since AlphaFold3 knows better part of this data due to training (Appendix A.2).

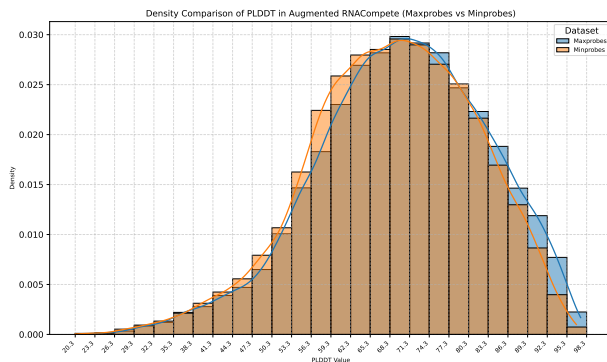


Figure 4.1: pLDDT comparison plot for RNACompete dataset

4.1.3 Counting RNA and protein most frequent trimers

Analysis of trimer frequencies revealed that protein sequences in positives and negatives exhibit similar distributions, suggesting protein sequence alone may not drive RNA recognition. In contrast, RNA trimer distributions differed substantially, with certain sequences (e.g., poly U) being significantly enriched in positive interactions. (Figure 4.4)

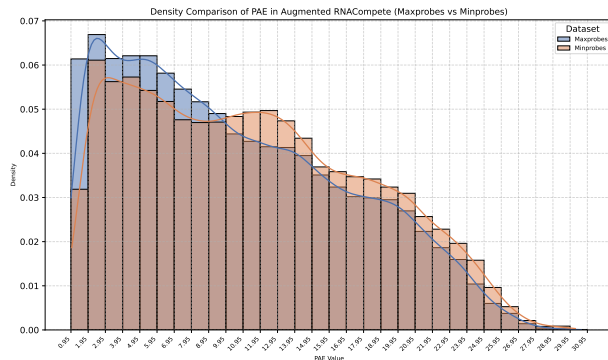


Figure 4.2: PAE comparison plot for RNACompete dataset

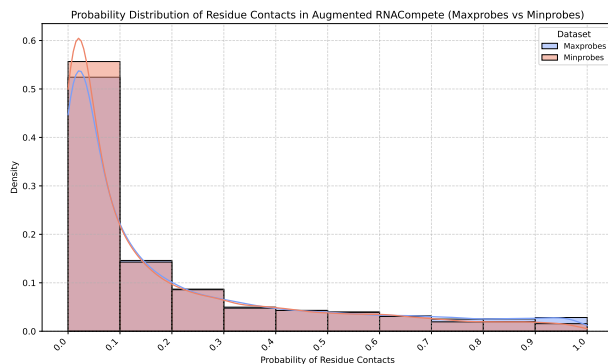


Figure 4.3: Probability of contacts comparison plot for RNACompete dataset

Also, a small comparison of frequent trimer frequencies in RNACompete proteins and a reference set of RNA-binding proteins from UniProtKB revealed minimal overlap. The most frequent trimers in RNACompete differed significantly from those in UniProtKB. This means RNACompete is not enriched in human specific sequence motifs. (Figure 4.5)

4.1.4 AlphaFold3 Confidence in Protein-to-RNA vs. RNA-to-Protein Modeling

Analysis of PAE(res, nt) and PAE(nt, res) distributions, which measure the predicted alignment error (PAE) of RNA from the protein's perspective and vice versa, revealed no consistent bias in AlphaFold3's confidence. Instead,

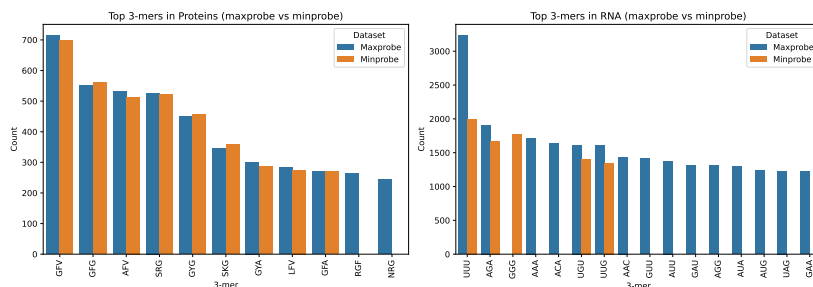


Figure 4.4: Frequent trimers RNACompete dataset

AlphaFold3’s uncertainty fluctuates across different PAE value ranges. (Figure 4.6)

4.1.5 AlphaFold3 Confidence Scores Across Interaction Distances

AlphaFold3 assigns a wide range of pLDDT and PAE scores across all interaction distances, indicating no strict bias toward short- or long-range RNA-protein interactions (Figure 4.7). The contours represent the data density. Inner contours represent regions of higher density.

4.1.6 AlphaFold3 Structural Confidence is Skewed Toward Proteins

Plotting the three pLDDT values for the trimer reveals a significant disparity between RNA and protein structural confidence in AlphaFold3 predictions. While protein pLDDT values are consistently high (70-100) with peaks around 85-97 for both interacting (maxprobes) and non-interacting (minprobes) pairs, RNA pLDDT values span a much wider range (20-100). The peak confidence for RNA structures is notably lower (40-60 for maxprobes, 30-70 for minprobes), indicating overall lower model certainty in RNA structural predictions. These findings suggest that AlphaFold3 is highly confident in modeling protein structures, but its reliability in RNA structure prediction remains limited, even in confirmed interacting cases. (Figure 4.8)

This tendency was also present when aggregating all pLDDT’s in the trimer by taking their mean. (Figure 4.9)

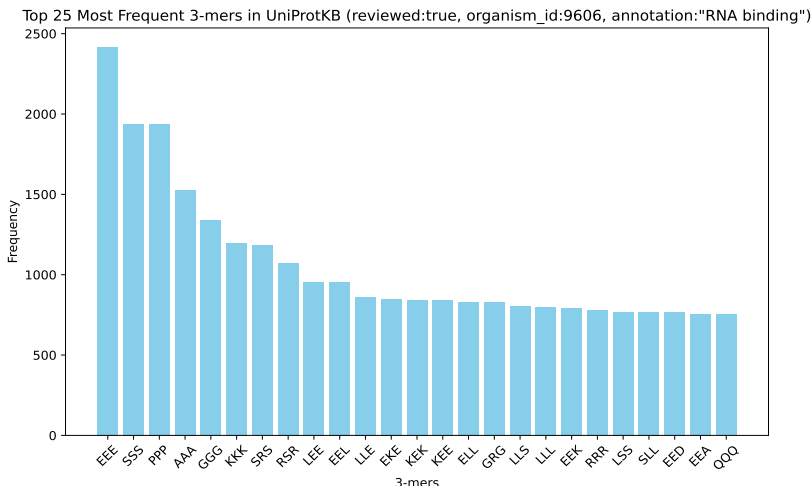


Figure 4.5: Frequent trimers UniProtKB dataset

4.1.7 Sequence Pattern Disparity Between RNACompete and AlphaFold3 Training Data

A comparison of trimer frequencies revealed minimal overlap between RNACompete-derived sequences (Maxprobe/Minprobe) and AlphaFold3 training data. The most frequent trimers in AlphaFold3 training set were largely absent in RNACompete, with no occurrences in Maxprobe nor Minprobe. Conversely, the most common RNACompete trimers appeared only rarely in AlphaFold3 training data. This suggests that AlphaFold3 was trained on a distinct sequence distribution, potentially impacting its ability to generalize to RNACompete-like RPIs. Additionally, the lack of overlap indicates no strong evidence of direct memorization of RNACompete sequences in AlphaFold3’s training. (Figure 4.10 - 4.12)

4.1.8 Confidence Comparison Between RNACompete and AlphaFold3 Predictions on its Training Set

The pLDDT distributions show that AlphaFold3 training data predominantly falls within 80–100, with a sharp peak at 95, while RNACompete examples span a broader range (50–95) with a peak around 70. (Figure 4.13)

Similarly, PAE scores for AlphaFold3 training data are concentrated in

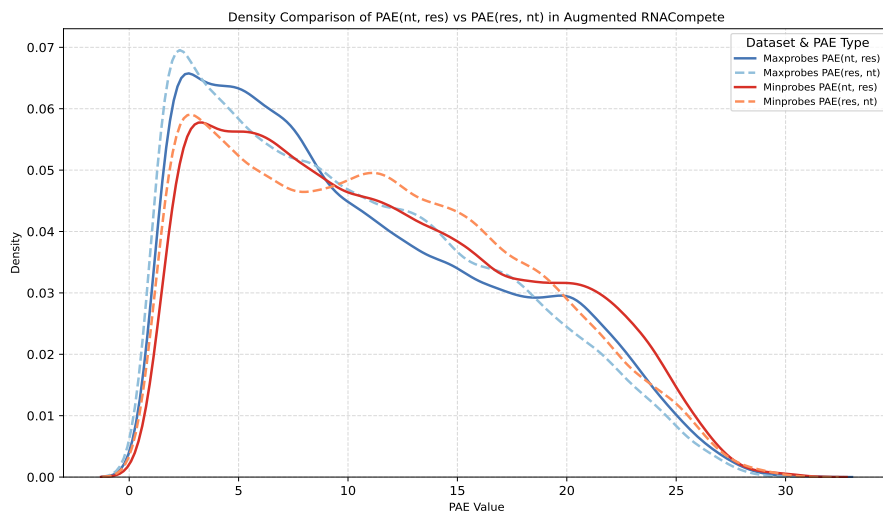


Figure 4.6: PAE from different perspectives comparison plot in RNACompete dataset

the (0,5) range, whereas RNACompete covers a much wider interval (0–30). (Figure 4.14)

The scatter plot further highlights this trend, with AlphaFold3 predictions on its training set clustering around high pLDDT values (65–98), while RNACompete covers a more diverse range (30–95). This suggest that AlphaFold3 may be overconfident in the predictions on its training set and less effective at modeling diverse RNA-protein interactions (like RNACompete), potentially favoring high-confidence structures seen during training (Figure 4.15). Contours again denote data density.

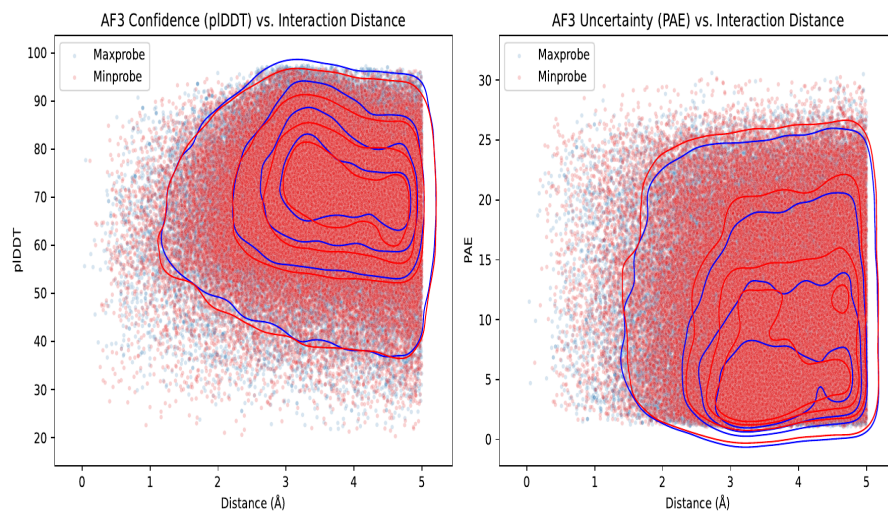


Figure 4.7: Confidence metrics and distances comparison in RNACompete dataset

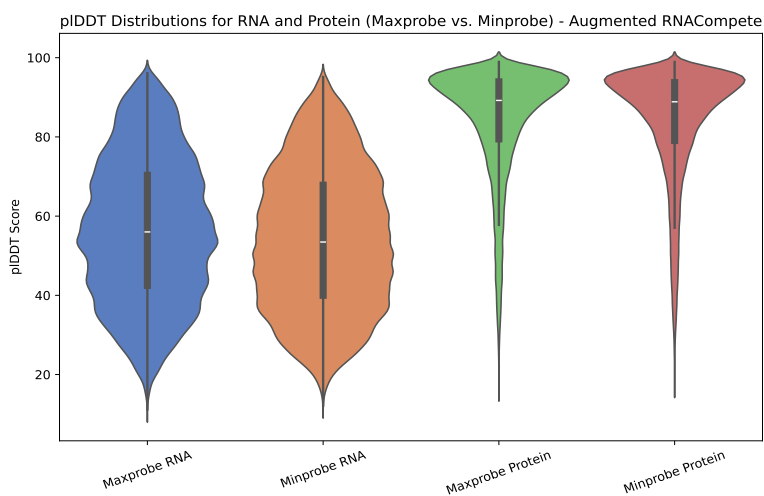


Figure 4.8: Trimers pLDDTs distributions in RNACompete dataset

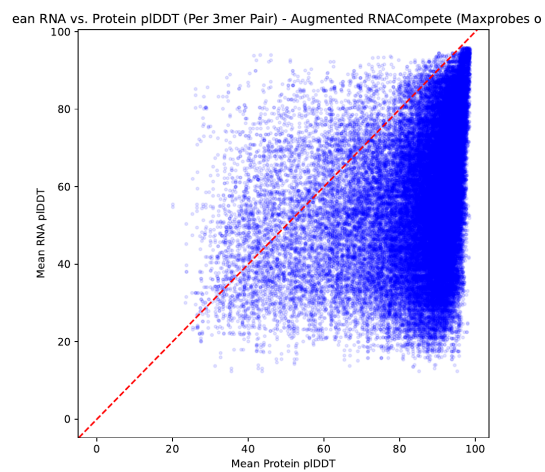


Figure 4.9: Trimers aggregated by taking the mean pLDDTs in RNACompete dataset

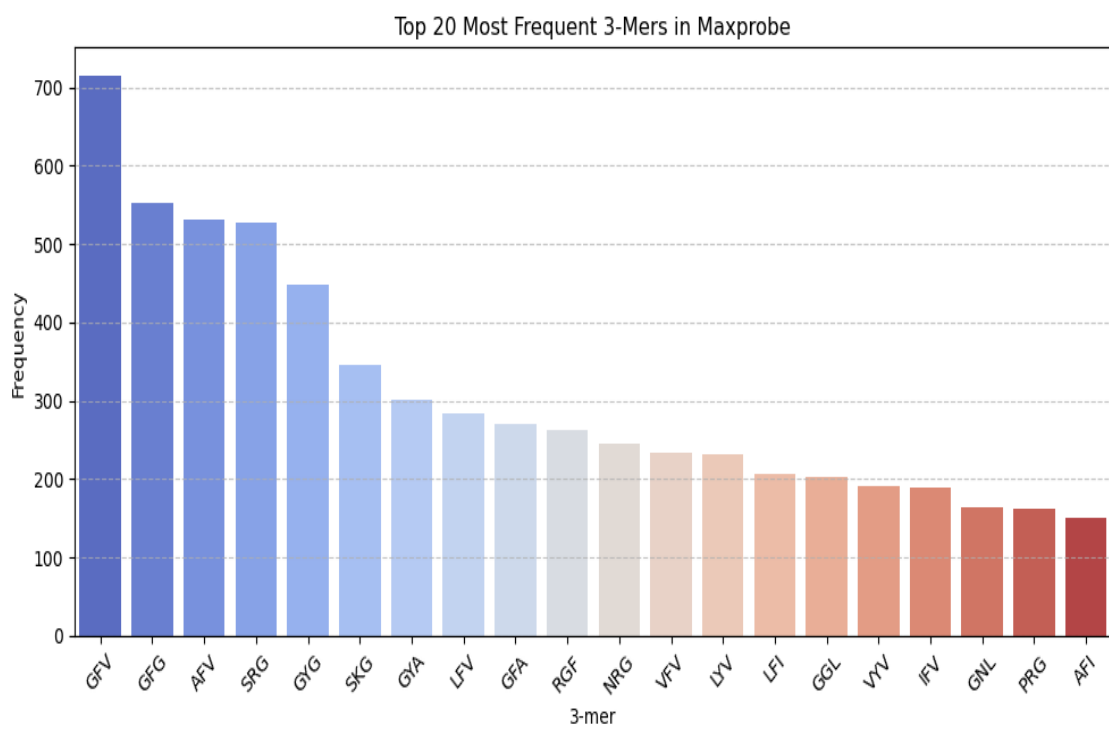


Figure 4.10: Top Protein trimers in RNACompete positives

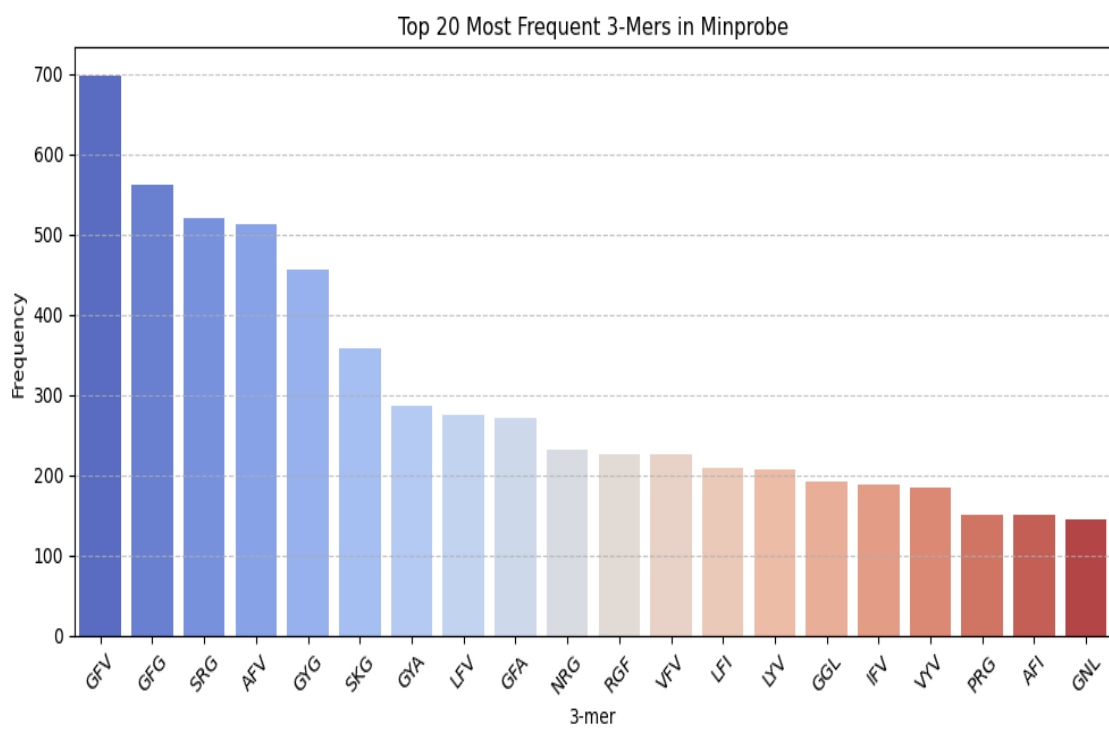


Figure 4.11: Top Protein trimers in RNACompete negatives

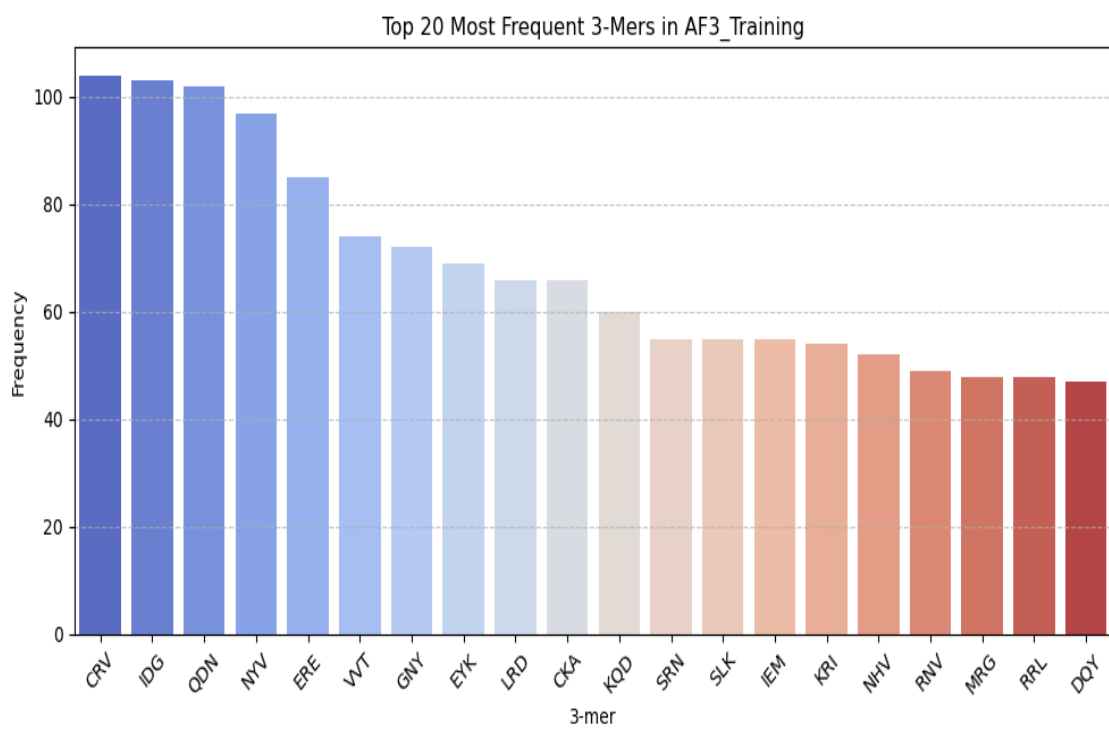


Figure 4.12: Top Protein trimers in AlphaFold3 training dataset

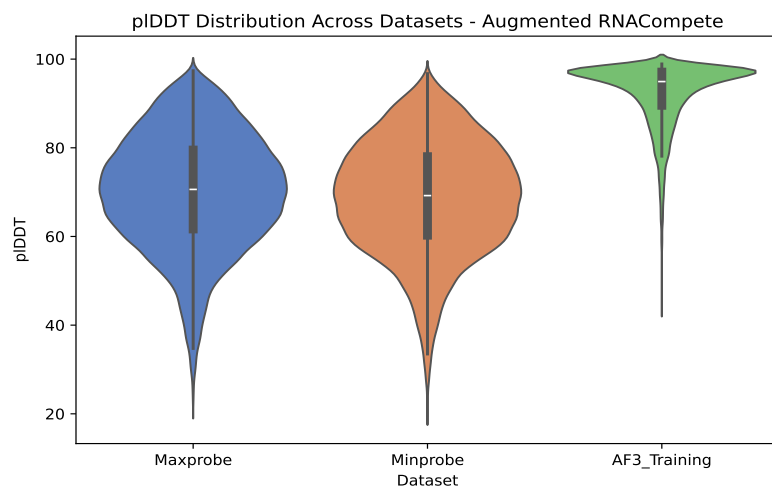


Figure 4.13: pLDDT comparison between RNACompete positives, negatives, and AlphaFold3 training set

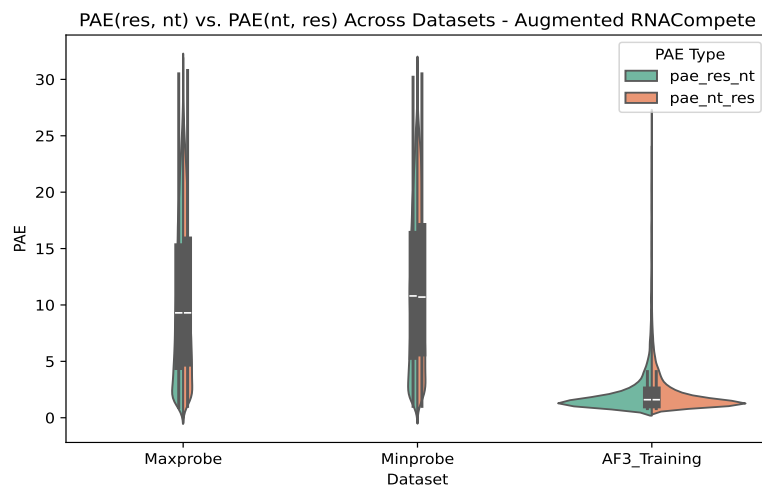


Figure 4.14: PAE comparison between RNACompete positives, negatives, and AlphaFold3 training set

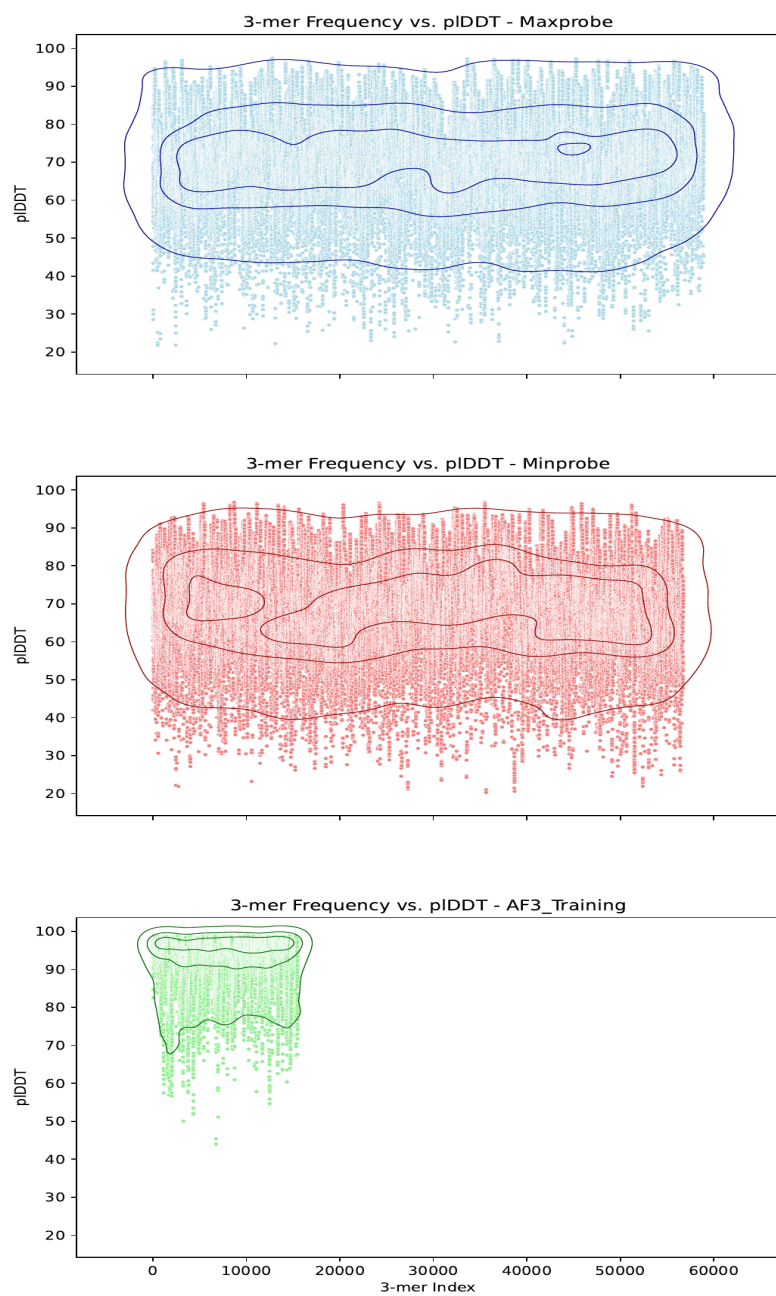


Figure 4.15: All pLDDTs for every trimer, for RNACompete positives, negatives and AlphaFold3 training set

Chapter 5

Classifying RNA-Protein Interactions with TabPFN and AlphaFold3

We used TabPFN seeking to evaluate whether AlphaFold3’s structural confidence metrics—including pLDDT, PAE and contact probabilities—, among other structural features, provided or not highly meaningful signals for RPIs.

To prepare features for every RNA-protein pair example, structural information of both sequences as well as AlphaFold3 confidence metrics were regarded.

Specifically the set of features was the following:

- RNA and protein sequences
- RPIEmbeddor prediction
- mean, standard deviation and median of pLDDT, PAE, distance in Angstroms of close residues, contact probabilities of close residues
- fraction disordered of the complex
- clashing percentage of the complex
- ipTM of the complex
- pTM of the complex
- ranking score of the complex relative to other prediction instances

- minimum PAE found between close residues
- RNA entropy

This data was then splitted into training set and testing set, ensuring both sets were disjoint, to avoid memorization during training.

In order to use TabPFN, a quick training must be done with a dataset from the task of interest.

After that the prediction performance (accuracy) in the testing set was 77.32 %

5.1 Feature importance with TabPFN's SHAP interpretability module

TabPFN contains an interpretability module from the SHAP (SHapley Additive exPlanations) library for machine learning interpretability in Python [3], which helps to explain the data with the help of SHAP values denoting the feature importances of the model.

The feature importance is shown in Figure 5.1:

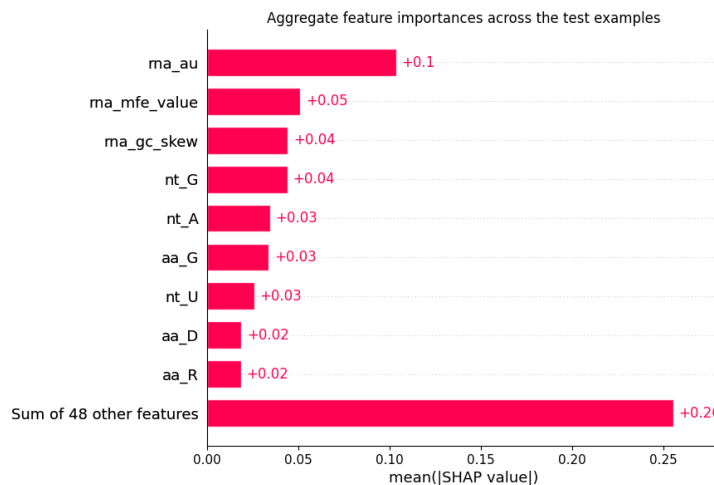


Figure 5.1: Most important features for prediction

As the ranking shows, structural information regarding how frequent are certain aminoacids an nucleotides within the sequences, stands out from the

rest of features. This suggests than AlphaFold3 confidence metrics (pLDDT, PAR, probability of residue-nucleotide contacts) might be not as meaningful to separate negative from positive examples in RPI classification.

Figure 5.2, is another plot produced by this interpretability module. In this plot, more details can be observed regarding how the values of the features actually affect the prediction:

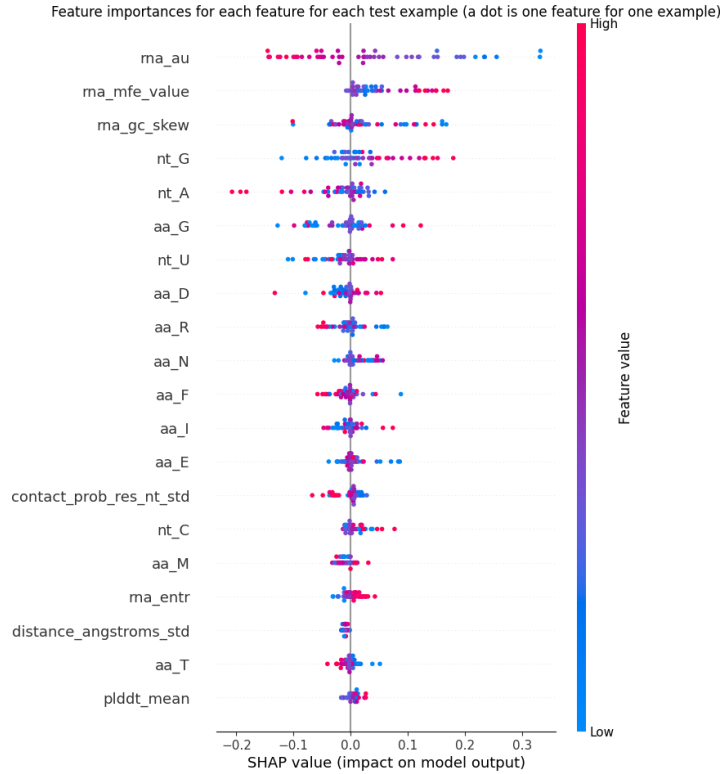


Figure 5.2: Important features with behavior regarding value variation

This plot provides a more detailed look at how each feature affects predictions. Each dot represents a test example, with its SHAP value on the x-axis.

Dots further to the right increase the prediction probability. Whereas dots to the left do the opposite. The color represent the values the feature can take. Therefore, when a fading color pattern can be seen among the dots of a feature, it implies that the value of the feature affects proportionally the prediction.

Chapter 6

Conclusion

Given the performance of the previous deep learning models, it becomes evident that identifying RPIs remains a challenging and complex task. Even specialized solutions like RPIembeddor can struggle at performing decently without being trained properly on RPI samples of the same nature than the ones we work with.

AlphaFold3 showed no major differentiation in some of its confidence metrics distributions between positive and negative RPIs. This confirms, for example, that PAE is not a strong standalone classifier for RPI. Still AlphaFold3 structural modeling was able to identify potential points within the sequences which could indicate binding regions. Even with this capabilities, more sophisticated adaptations are needed for it to identify actual RPIs.

TabPFN demonstrated a great predictive power given the right set of features. This means that such a tool could be a great candidate to consider for more effective model pipelines to address this complicated task.

Appendix A

Appendix

A.1 Preprocessing script of RNA sequences for RPIembeddor

RNA sequences needed to be specified within a FASTA file.

The functionalities of this script were:

- Mode 1: taking the whole RNA sequence: No subsequence was extracted in this mode.

In the description line of the RNA sequence, no parameter is appended to the sequence ID.

For example: from the RNA sequence "seq1", take the whole sequence. Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1
AGCGAUCCGG
```

The script would take the complete RNA sequence as a result.

- Mode 2: extracting a subsequence from the RNA sequence: Here a start and end index were required.

Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the interval to extract is appended to the sequence ID, with the format:

```
| start_index=1 end_index=5
```

Note that `start_index` is a 0-based index while `end_index` is a 1-based index (this is how “bedtools” works).

For example: from the RNA sequence “seq1”, extract the subsequence from the third character to the seventh character, both inclusive. Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1 | start_index=2 end_index=7
AGCGAUCCGG
```

The script would take the subsequence CGAUC from the RNA sequence as a result.

- Mode 3: extracting subsequences of iteratively incrementing lengths from a fixed starting end or interval of the RNA sequence. This includes three submodes:

- Submode 1:

Subsequences extracted from the sequence left end.

Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the submode of operation and the spacing are appended to the sequence ID, with the format:

```
| submode=0 spacing=2
```

For example: from the RNA sequence “seq1”, extract the subsequences with length iteratively increasing by two nucleotides from the left end of the sequence. Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1 | submode=0 spacing=2
AGCGAUCCGG
```

The script would take the subsequences AG, AGCG, AGCGAU, AGCGAUCC and AGCGAUCCGG from the RNA sequence as a result.

– Submode 2:

Subsequences extracted from the sequence right end.

Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the submode of operation and the spacing are appended to the sequence ID, with the format:

```
| submode=1 spacing=3
```

For example: from the RNA sequence "seq1", take the subsequences with length iteratively increasing by three nucleotides from the right end of the sequence. Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1 | submode=1 spacing=3  
AGAUCCCGAUCCGGCCG
```

The script would take the subsequences CCG, CGGCCG, AUC-CGGCCG, CCGAUCCGGCCG and AUCCCGAUCCGGCCG from the RNA sequence as a result.

– Submode 3:

Subsequences extracted from some middle interval within the sequence. The increments would be done simultaneously to the left and to the right of the specified fixed starting interval until reaching, if possible, the half of the sequence length at each direction.

Parameters needed in the FASTA file entry. In the description line of the RNA sequence, the submode of operation, the spacing and the interval left and right limits are appended to the sequence ID, with the format:

```
| submode=2 spacing=2 middle_start_left=5  
middle_start_right=7
```

Note that middle_start_left is a 0-based index while middle_start_right is a 1-based index (this is how "bedtools" works).

For example: from the RNA sequence "seq1", extract the subsequences with length iteratively increasing by one nucleotide to each side of the interval starting with the third nucleotide "A" and ending with the seventh nucleotide "U". Then in the FASTA file, the corresponding RNA entry would be:

```
>seq1 | submode=2 spacing=1 middle_start_left=2  
      middle_start_right=7  
CGAGGCUGGGCGCGCCC
```

The script would extract the subsequences AGGCU, GAGGCUG, CGAGGCUGG, CGAGGCUGGG, CGAGGCUGGGC, CGAGGCUGGGCG. from the RNA sequence as a result.

More specifically what the scripts does is the following: From the maximum allowed length for extracted subsequences (let's suppose in this case this is fifteen), the script subtracts the interval length (five). This difference (ten) is taken as the maximum number of nucleotides to extend from. Then, the script extracts subsequences extending iteratively to both sides of the interval until reaching half of this difference (five) or some end of the sequence. With this, subsequences of diverse lengths are extracted and all of them include the interval of interest.

A.2 AlphaFold3 prediction distributions for alternative dataset

In this section, the plots for the alternative dataset of chapter three discussion are shown.

The overall distributions are also very similar for both positives and negatives RPI examples.

The only major difference with respect to the RNACompete dataset plots is the naturally better confidences for the positive set. AlphaFold3 knows better part of this data due to training.

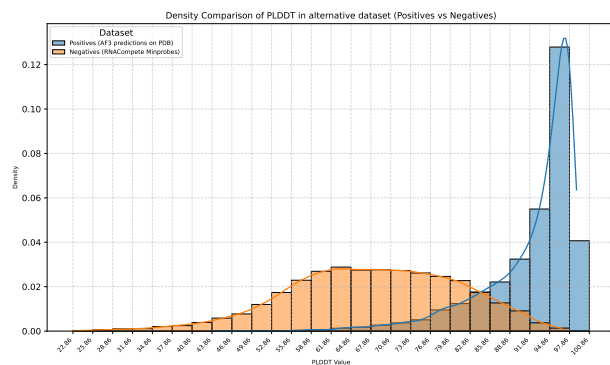


Figure A.1: pLDDT comparison plot for alternative dataset

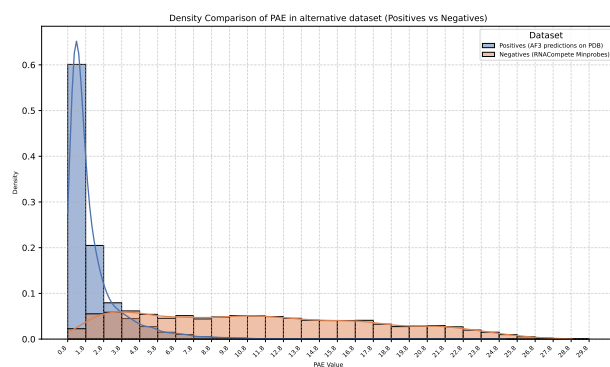


Figure A.2: PAE comparison plot for alternative dataset

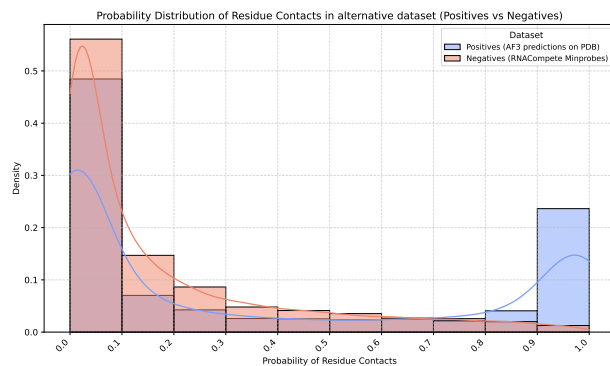


Figure A.3: Probability of contacts comparison plot for alternative dataset

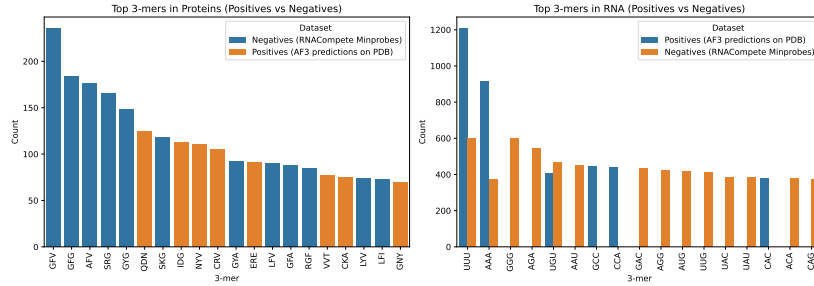


Figure A.4: Frequent trimers alternative dataset

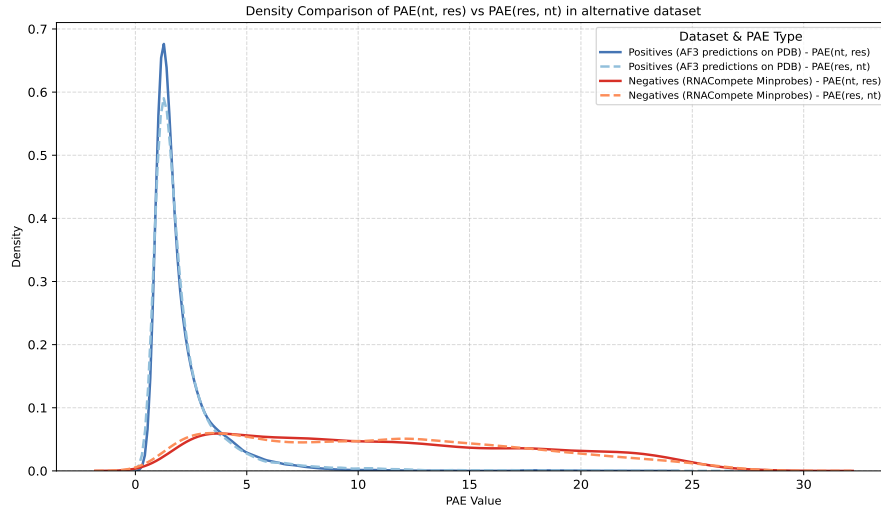


Figure A.5: PAE from different perspectives comparison plot in alternative dataset

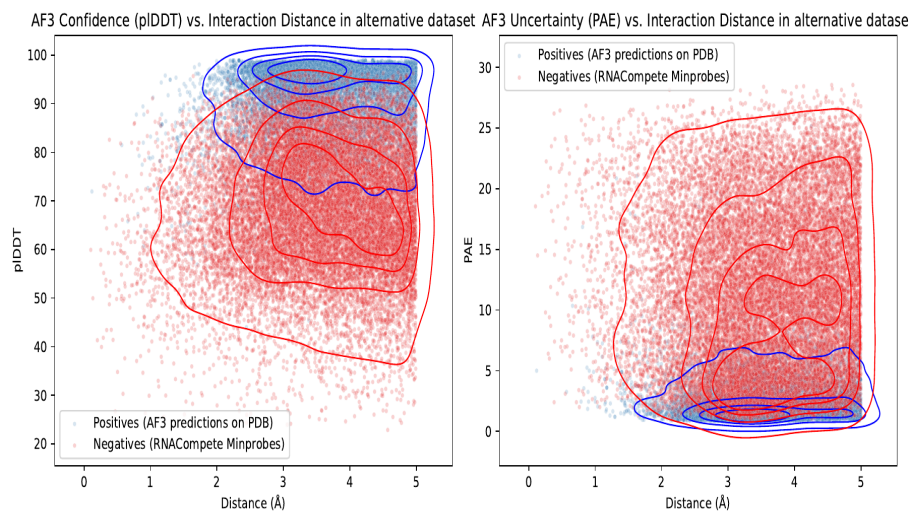


Figure A.6: Confidence metrics and distances comparison in alternative dataset

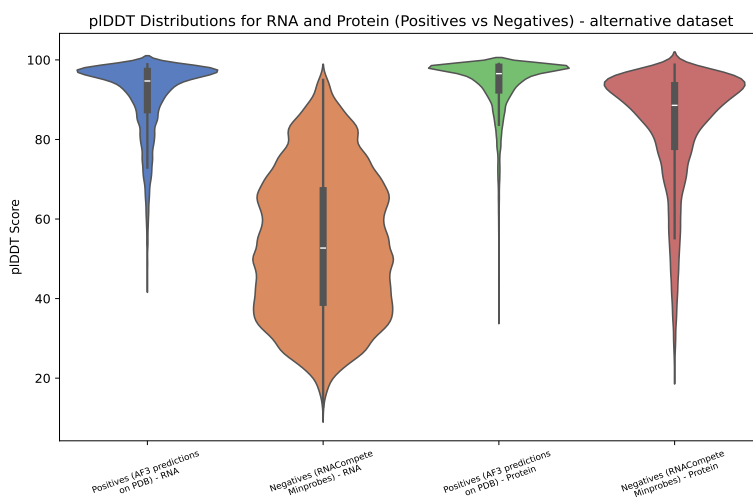


Figure A.7: Trimers pLDDTs distributions in alternative dataset

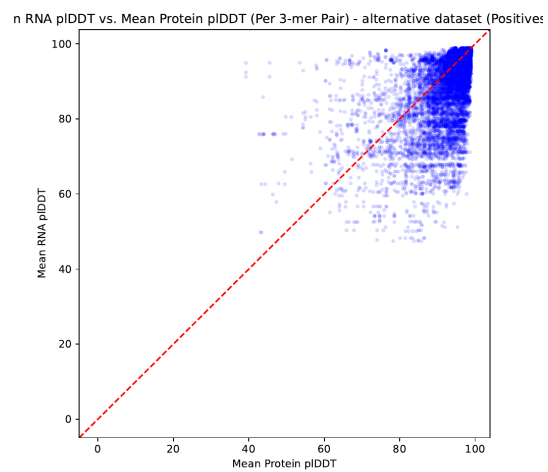


Figure A.8: 3-mer's aggregated by taking the mean pLDDTs in alternative dataset

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